

QUANG-HUY NGUYEN

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RESEARCH INTERESTS

Computational Biology: My work spans multi-omics studies integrating genomics, transcriptomics, metabolomics, and proteomics. This includes bulk, single-cell, and spatially resolved analyses, pathway analysis, variant calling (both germline and somatic), genome-wide association studies, and polygenic risk scores.

Computational Healthcare Informatics: I develop and apply data-driven methods for learning from heterogeneous clinical and biological data to support decision-making in medicine and public health.

I also address core AI challenges raised by these problems, including multi-modal modeling, agentic reasoning, uncertainty quantification, federated and collaborative learning, etc.

EDUCATION

Auburn University

Ph.D. in Computer Science & Software Engr - cGPA: 3.73/4.0

Auburn, AL

08/2023 – Present

Advisor: Professor Wei-Shinn Ku.

- Developed *FedeKD*, a federated knowledge distillation framework for robust knowledge transfer under heterogeneous settings. Achieved a significant reduction in negative transfer while improving predictive performance across 6 vision and medical imaging datasets.
- Developed *scBatchProx*, a federated learning-inspired framework for post-hoc single-cell batch integration via latent refinement, enabling continual and cumulative updates under dataset growth. Achieved Macro-F1 up to 0.76 with robustness to extreme batch imbalance.
- Developed *FedWQ-CP*, a federated conformal prediction framework for uncertainty quantification under joint data and model heterogeneity. Achieved robust client-level and global coverage across 7 vision and medical imaging datasets.

Dai Nam University

Bachelor of Pharmacy - cGPA: 8.09/10.00

Hanoi, Vietnam

10/2015 – 08/2020

RESEARCH EXPERIENCE

Department of Computer Science, HUST

Visiting Graduate Student

Hanoi, Vietnam

12/2024 – 01/2025

Advisor: Associate Professor Duc-Hau Le.

- Developed *PAG-Agent*, a web-based, LLM-driven platform for context-aware pathway analysis of bulk and single-cell RNA-seq data, unifying differential expression analysis, pathway enrichment, meta-analysis, literature-assisted interpretation, and hypothesis generation. Designed a novel context-aware pathway prioritization framework that incorporates experimental conditions and research objectives to improve the identification of biologically relevant pathways while reducing generic enrichment signals, a common limitation of conventional ORA-based methods.
- Developed *DeepSSC*, a semi-supervised multi-omics classifier for cancer subtype prediction. Achieved 96.0% accuracy (99.2% AUC) across 3 TCGA cohorts, with up to 19% improvement over graph-based baselines; mentored undergraduate students on model development.

Department of Computer Science and Software Engineering, Auburn University

Graduate Research Assistant

Auburn, AL

08/2023 – 08/2024

Advisor: Research Assistant Professor Phi Hung Bya and Associate Professor Tin Nguyen.

- Conducted a literature review on functional analysis methodologies in LC-MS-based targeted and untargeted metabolomics. Dockerized 13 surveyed software tools as services, simplifying installation and execution to enhance accessibility for readers with limited IT expertise.
- Conducted a literature review on large language models and their applications in biomedical domain.
- Led the *Candidozyma auris* project; performed BLAST-based homology mapping and KEGG/GO enrichment analyses to characterize pathways involved in stress adaptation and antifungal resistance; contributed to manuscript editing and proofreading.
- Led the type 2 diabetes (T2D) polygenic score project; developed a genotype-to-pathway analysis pipeline that derives gene-level reference-deviation signals directly from VCF genotypes without matched transcriptomic data, identified T2D-associated pathways, and demonstrated strong predictive performance of pathway-specific polygenic risk scores; co-drafted the manuscript.

School of Computer Science and Engineering, Thuyloi University

Visiting Scientist

Hanoi, Vietnam

10/2021 – 07/2023

Advisor: Associate Professor Duc-Hau Le.

- Developed *DrGA*, an R toolkit for automated analysis of cancer driver genes, unifying gene set enrichment, gene-clinical association, module-level analysis, and patient stratification workflows.

School of Computer Science and Engineering, Thuyloi University

Visiting Student

Hanoi, Vietnam

12/2017 – 07/2019

Advisor: Associate Professor Duc-Hau Le.

- Read and replicated research papers, designed experiments and conducted algorithm on high-dimensional omics data in R.

INDUSTRY EXPERIENCE

Center for Biomedical Informatics, Vingroup Big Data Institute

Research Scientist

Hanoi, Vietnam

10/2020 – 09/2021

Managers: Associate Professor Duc-Hau Le and Dr. Nam Sy Vo.

- Developed *oCEM*, an R package for detecting overlapping co-expressed gene modules using decomposition methods (ICA/IPCA), with a permutation-based approach for selecting the optimal number of components.
- Designed and implemented a multi-omics analysis pipeline in R that detected novel prognostic subgroups of breast cancer, achieving a Cox log-rank p-value of 0.002 compared to the PAM50 classification scheme's p-value of 0.598. Led manuscript preparation and writing.
- Co-mentored an undergraduate student in clustering uveal melanoma patients in R.
- Contributed to the design and planning of the VN1K variant detection pipeline for large-scale genomic analysis of the Vietnamese population. Contributed to manuscript proofreading.

Research Intern

03/2020 – 10/2020

Internship mentor: Associate Professor Duc-Hau Le.

- Developed and implemented a pipeline in R to systematically identify and analyze cancer driver genes using multi-omics data, ensuring a consistent approach to uncover key genetic factors contributing to cancer progression.

RESEARCH PROJECTS AND COMPETITIONS

nano-scFMs (🔗 Repo)

Independent Open-Source Project

Auburn, AL

06/2026 – Present

- Built a collection of lightweight, faithful PyTorch reimplementations of single-cell foundation models to improve readability, reproducibility, and accessibility for AI biology research.
- Reimplemented model architectures, preprocessing pipelines, tokenization workflows, pretrained checkpoint loading, and embedding extraction for single-cell `.h5ad` data.

Cell Maps for AI (CM4AI) CodeFest, UAB

NIH Bridge2AI Consortium

Birmingham, AL

08/2025

- Collaborated in a team of 5 to create separate embeddings of yeast fitness data, proteomic mass spectrometry, and cellular imaging, then fuse them into a co-embedding to construct a with-yeast multiscale subcellular hierarchy that uncovered hidden biological processes in triple-negative breast cancer.
- Implemented the analysis pipeline based on the agreed workflow in Python.

The 1000 Vietnamese Genomes Project (VN1K)

Vingroup Joint Stock Company

Hanoi, Vietnam

2019 – 2024

PI: Associate Professor Duc-Hau Le and Dr. Nam Sy Vo.

GenomeAsia100K

NTU and others

Hanoi, Vietnam

2017 – 2022

PI: Dr. Stephan Schuster (Singapore side) and Associate Professor Duc-Hau Le (Vietnam side).

TEACHING EXPERIENCE

Department of Computer Science and Software Engineering, Auburn University

Graduate Teaching Assistant for COMP-1220 Python & COMP-1230 Matlab

Auburn, AL

08/2024 – 05/2026

- Teaching Python and Matlab fundamentals to ≈ 60 undergraduates and guiding them through problem-solving exercises in lab sections.

PUBLICATIONS

†co-first author, *co-corresponding author

Preprints & Preparations

5. **Quang-Huy Nguyen**, Jiaqi Wang*, Wei-Shinn Ku*. *FedeKD: Energy-Based Gating for Robust Federated Knowledge Distillation under Heterogeneous Settings*. **arXiv**, 2605.05553. ([Under review](#)). 2026
4. **Quang-Huy Nguyen**, Jiaqi Wang*, Wei-Shinn Ku*. *scBatchProx: Federated-Inspired Refinement for Stable Cell-Type Discriminability under Heterogeneous Batch Compositions*. **arXiv**, 2602.00423. ([Under review](#)). 2026
3. **Quang-Huy Nguyen**, Jiaqi Wang*, Wei-Shinn Ku*. *Conformalized Neural Networks for Federated Uncertainty Quantification under Dual Heterogeneity*. **arXiv**, 2602.23296. ([Under review](#)). 2026
2. Hoang Le†, Van-Minh Nguyen†, Duc-Hai Tran, Van-Anh Tran, Van-Nui Nguyen, **Quang-Huy Nguyen**†*, Duc-Hau Le*. *A Semi-Supervised Deep Learning Framework for Multi-Omics Cancer Subtype Classification and Biomarker Discovery*. **bioRxiv**, 2022.01.13.476268. ([Under review](#)). 2026
1. **Quang-Huy Nguyen**, Zeru Zhang, Duc-Hau Le*, Jake Y. Chen*, Wei-Shinn Ku*, Hao Chen*, Zongliang Yue*. *PAG-Agent: a biologist-oriented research assistant for context-aware pathway-level analysis and interpretation*. **bioRxiv**, 2026.06.02.729674. ([Under review](#)). 2026

Peer-reviewed publications

10. Trang T. H. Tran, ..., **Quang-Huy Nguyen**, ..., Quan Nguyen*, Van H. Vu*, Nam S. Vo*. *VN1K: a genome graph-based and function-driven multi-omics and phenomics resource for the Vietnamese population*. **bioRxiv**, 2025.04.15.648991. **Nature Communications** (Acceptance in Principle). 2026
9. Ching-Lan Chang†, Michael A. Mosh†, **Quang-Huy Nguyen**†, ..., David Hess*, Edwin C. Oh*. *Wastewater intelligence predicts the emergence of clinically-relevant and drug-resistant *Candidozyma auris* at healthcare facilities*. **Nature Communications**. 2026
8. **Quang-Huy Nguyen**, Ha Nguyen, Edwin C. Oh, Tin Nguyen. *Current approaches and outstanding challenges of functional annotation of metabolites: a comprehensive review*. **Briefings in Bioinformatics**, 25(6):bbae498. 2024
7. **Quang-Huy Nguyen**, Tin Nguyen, Duc-Hau Le. *DrGA: cancer driver gene analysis in a simpler manner*. **BMC Bioinformatics**, 23:86. 2022
6. **Quang-Huy Nguyen**, Tin Nguyen, Duc-Hau Le. *Identification and validation of a novel three hub long noncoding RNAs with m6A modification signature in low-grade gliomas*. **Frontiers in Molecular Biosciences**, 9:801931. 2022
5. **Quang-Huy Nguyen**, Duc-Hau Le. *oCEM: Automatic detection and analysis of overlapping co-expressed gene modules*. **BMC Genomics**, 23:39. 2022
4. Thi Hai Yen Nguyen†, Tin Nguyen, **Quang-Huy Nguyen**†*, Duc-Hau Le*. *Re-identification of patient subgroups in Uveal Melanoma*. **Frontiers in Oncology**, 11:731548. 2021
3. **Quang-Huy Nguyen**, Duc-Hau Le. *Similarity calculation, enrichment analysis, and ontology visualization of biomedical ontologies using UFO*. **Current Protocols in Bioinformatics**, 1:e115. 2021
2. **Quang-Huy Nguyen**, Duc-Hau Le. *Improving existing analysis pipeline to identify and analyze cancer driver genes using multi-omics data*. **Scientific Reports**, 10:20521. 2020
1. **Quang-Huy Nguyen**, Hung Nguyen, Tin Nguyen, Duc-Hau Le. *Multi-omics analysis detects novel prognostic subgroups of breast cancer*. **Frontiers in Genetics**, 11:574661. 2020

PEER-REVIEWED CONFERENCES

1. Hung Nguyen, Bang Tran, Duc Tran, **Quang-Huy Nguyen**, Duc-Hau Le, Tin Nguyen. *Disease subtyping using community detection from consensus networks*. **2020 12th International Conference on Knowledge and Systems Engineering (KSE)**, Can Tho, Vietnam, IEEE, 2020; 318–323. Oral Presentation.

PRESENTATIONS AND TALKS

1. 2024 MW CTR-IN Annual Meeting at The Westin Las Vegas Hotel & Spa, Las Vegas, NV. *Topic: A Systems-level Approach for Predicting Patient Survival using Multi-omics Data*. 2024

HONORS AND AWARDS

Benchmarks Resource Grant Program

Auburn, AL

Kaggle Research Grant Program

07/2026

- Selected for the Benchmarks Resource Grant Program of the Kaggle Research Grant Program to develop an open-source benchmark for evaluating LLMs on context-aware biological pathway prioritization, receiving a $5\times$ increase in AI Models quotas (from \$10 to \$50 daily and from \$100 to \$500 monthly).

Gold Reviewer Award

Auburn, AL

Issued by ICML 2026

05/2026

- Gold Reviewer Award (top 25% reviewers) at ICML 2026.

Google PhD Fellowship University-Level Nomination

Auburn, AL

Associated with Auburn University

04/2026

- One of three graduate students nominated university-wide by AU Graduate School for 2026 Google PhD Fellowship.

Certificate of Merit for Academic Excellence and Class Leadership

Hanoi, Vietnam

Associated with Dai Nam University

08/2020

- Top 3% of the Department's 2015 cohort (nearly 110 students).
- Class president in 5 consecutive years (2015-2020).

27th Mathematics Olympic for Students

Nha Trang, Vietnam

Issued by Vietnam Mathematical Association

04/2019

- 🏆 **Received an honorable mention** in Linear Algebra among 640 undergraduates representing 80 universities, academies, colleges across Vietnam.
- Taught myself linear algebra from scratch in preparation for the competition.
- Proudly attended the competition as the first and only representative of my university and became the first Vietnamese non-engineering student in the history of the competition.

TECHNICAL SKILLS

• **Programming:** Python, R, Matlab, JavaScript

• **Single-Cell & Omics:** Scanpy, AnnData, MuData, Seurat

• **Web Development:** HTML, CSS, RESTful APIs

• **Deep Learning & ML:** PyTorch, TensorFlow 2, scikit-learn

• **Database:** MySQL, SQLite, Oracle, MongoDB

• **Clouds & Tools:** BigQuery, AWS, Git, GitHub, Docker